

PhysicsMind: Sim and Real Mechanics Benchmarking for Physical Reasoning and Prediction in Foundational VLMs and World Models

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Abstract

Modern foundational Multimodal Large Language Models (MLLMs) and video world models have advanced significantly in mathematical, common-sense, and visual reasoning, but their grasp of underlying physics remains underexplored. Existing benchmarks for this question often rely on synthetic Visual Question Answering templates or emphasize perceptual video quality, which is only indirectly related to whether outputs obey physical laws. To address this gap, we introduce **PhysicsMind**, a controlled mechanics benchmark spanning both real-world and simulated settings across three canonical mechanics scenarios: Center of Mass, Lever Equilibrium, and Newton’s First Law. PhysicsMind includes two tasks: (i) VQA, which tests whether models can infer physically relevant quantities and outcomes from images or short videos, and (ii) Video Generation (VG), which evaluates whether predicted motion trajectories remain physically consistent with center-of-mass, torque, and inertial constraints. This design provides controlled mechanics diagnostics across perception and prediction, while enabling scenario-dependent sim-to-real analyses when real and simulated settings are directly comparable. Evaluations of recent multimodal and video generation models show persistent reliance on appearance heuristics and frequent violations of basic mechanics. These results indicate that current scaling and training remain insufficient for robust physical reasoning and physical consistency, underscoring PhysicsMind as a controlled mechanics benchmark for model diagnostics. An anonymized repository is available at <https://anonymous.4open.science/r/PhysicsMind-C4C8>;

CCS Concepts

• **Computing methodologies** → **Computer vision tasks**; *Visual content-based indexing and retrieval*; • **Applied computing** → *Physics*.

Keywords

physical reasoning, multimodal large language models, video generation, benchmark, world models

1 Introduction

Recent multimodal and video generation models have advanced rapidly in mathematical and commonsense reasoning [1, 32, 39], yet their grasp of real-world physical dynamics remains limited [11, 29]. Unlike symbolic tasks, physical reasoning requires grounding abstract concepts (force, mass, and inertia) into visually observable and temporally coherent motion. As mechanical laws govern how objects interact and remain stable, they are central to perception and embodied decision-making, making them a critical benchmark for current Visual-Language Models (VLMs) and Video Generation Models (VGMs).

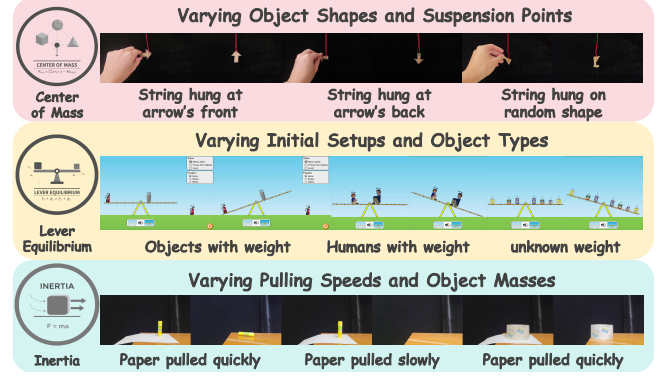


Figure 1: Three representative mechanics scenarios in PhysicsMind, instantiated in controlled tabletop and simulated settings for controlled mechanics diagnostics.

Prior work on evaluating physical reasoning in generative models falls primarily into two large categories. The first line studies VLMs on physical question answering and visual reasoning, such as IntPhys [36], CRAFT [5], CausalVQA [14], MVP [25], PhysBench [11], and PhysUniBench [41]. These benchmarks provide short videos or images paired with template questions about collisions, stability, and causal relations. While they are valuable for probing whether models can answer physics questions, the visual scenarios are often synthetic, with regular question formats highly regular, and plain evaluation methods, allowing methods to exploit linguistic or visual shortcuts without interpreting and reasoning about mechanical interactions.

The second category evaluates VGMs and World Models on their abilities to produce physically consistent dynamics. Recent benchmarks such as VBench [22], PhyGenBench [28], Physics-IQ [29], MORPHEUS [46], WorldModelBench [26], and WorldScore [13] are widely used to test diffusion models like Sora and Veo. However, these methods suffer from simple evaluation protocols focusing on pixel-level reconstruction, motion smoothness, or human-judged plausibility. Visually impressive sequences can still violate basic mechanics, exhibit unstable or energy non-conserving motion, or hallucinate impossible trajectories.

To move beyond this fragmented landscape, we introduce PhysicsMind, a controlled mechanics benchmark that evaluates both VQA and video generation on controlled real and simulated data. It focuses on three textbook scenarios (Figure 1) so that errors can be traced to specific laws, while matched real/sim subsets support targeted transfer analysis rather than broad sim-to-real claims.

We extensively evaluate state-of-the-art VLMs and video generators. Despite strong perceptual capabilities and high visual fidelity, current baselines show persistent gaps in basic mechanics: VLMs

Table 1: Comparison of physics-oriented VQA benchmarks. Columns summarize dataset realism, task modality, annotation format, and the scope of physical evaluation; PhysicsMind adds mechanics-specific cross-checks across controlled mechanics scenarios in both real and simulated settings.

Benchmark	Models	Realism	Task Modal	Answer Type	Video Length	Max Objects	Eval Aspects	CrossCheck
IntPhys [36]	2	Sim	Dynamic	Score	7s	3	3	No
CRAFT [5]	9	Sim	Dynamic	Choice	10s	6+	3	No
CausalVQA [14]	6	Real	Dynamic	Choice	24s	3	5	No
MVP [25]	13	Real	Dynamic	Choice	9s	6+	5	No
PhysBench [11]	39	Real+Sim	Dynamic+Static	Choice	8 frames	5	4	Yes
PhysicsMind (Ours)	22	Real+Sim	Dynamic+Static	Choice	4–11s	6+	6	Yes

Table 2: Comparison of video generation and world-model benchmarks. The table contrasts realism, supervision, and evaluation scope; PhysicsMind provides controlled mechanics cross-checks for mechanics-focused video prediction in both real and simulated settings, with scenario-specific sim-to-real comparability.

Benchmark	Realism	Ground-Truth Video	Video Length (s)	Input Modalities	Max Objects	Video Dim	Eval Aspects	CrossCheck
WorldModelBench [26]	Sim	No	2–5	Text/Image	1	3	7	No
MORPHEUS [46]	Real	No	2–4	Image/Video	2	3	3	No
WorldScore [13]	Real	yes	2–3	Image	6+	3	10	No
PhyGenBench [28]	Sim	No	3–11	Text	1	3	4	No
VBench [22]	Sim	No	2	Text	3	3	16	No
Physics-IQ [29]	Real+Sim	yes	8	Image/Video	3	3	4	No
PhysicsMind (Ours)	Real+Sim	Yes	5–10	Image/Video	6+	2/3	8	Yes

rely on appearance heuristics and struggle with counterfactual or logically linked questions, while video generators frequently produce visually plausible but physically inconsistent dynamics. Our main contributions are as follows.

- (1) We introduce PhysicsMind, a controlled mechanics benchmark with aligned real/sim splits for scenario-level diagnostics.
- (2) We design mechanics-specific evaluation protocols that go beyond overall accuracy or visual quality: for VQA, scenario-specific subtasks (e.g., position vs. rotation, equilibrium vs. adjustment), and for video generation, physical-consistency metrics for center-of-mass alignment, lever final-state correctness, and trajectory, speed, and acceleration consistency.
- (3) We conduct a systematic study of modern VLMs and video generators on PhysicsMind, revealing consistent failure modes such as reliance on appearance cues, weak causal reasoning, and frequent violations of basic mechanics, thereby outlining concrete targets for future multimodal models with stronger physical reasoning and physical consistency.

2 Related Work

Physical reasoning benchmarks. Physical reasoning benchmarks aim to test whether a model can infer and apply basic physical principles from visual input. Early work relied on synthetic, highly controlled environments. CLEVR [24] targets compositional reasoning in rendered scenes, while IntPhys [36], CRAFT [5], and PHYRE [6] probe intuitive physics and causal interactions in simulators. More recent datasets, such as CausalVQA [14], MVP [25],

PhysBench [11], PhysUniBench [41], and Physics-IQ [29] move toward richer video question answering with more realistic footage. However, most benchmarks use templated questions or narrow scenarios [8, 17]. While large multimodal models [4, 12, 30, 39] have advanced language-conditioned reasoning, they often misinterpret basic mechanics (Table 1). We therefore focus on mechanics-specific VQA diagnostics under controlled setups.

Video generation and world models. World models [1, 9, 10, 32, 47] and video generators learn predictive dynamics from streams of observations. Classical models like PlaNet [19] and Dreamer [20] focus on control returns. Recent generative benchmarks, including VBench [22], PhyGenBench [28], Physics-IQ [29], MORPHEUS [46], WorldModelBench [26], WorldScore [13], WorldPrediction [7], and AutumnBench [42], assess perceptual quality which often misaligns with physical consistency [21, 27, 29]. Although modern video generators [15, 32, 40] excel at synthesis, they frequently produce physically inconsistent motion (Table 2), motivating our mechanics-specific motion diagnostics.

Position. Despite rapid progress in multimodal foundation models [4, 12, 30, 39] and video generators [15, 32, 40], fundamental failures in basic mechanics remain common across both simulation and real-world video settings. We therefore position our work as a controlled mechanics benchmark for targeted diagnostics and transfer analysis, rather than as a comprehensive test of physical intelligence. Recent studies also report recurring failure patterns—including object-state drift, force–motion mismatch, and temporal inconsistency—but their analyses are often qualitative and difficult to compare across settings [21, 27, 29]. We therefore emphasize unified, mechanics-specific metrics with matched simulated/real subsets, making these errors directly measurable and comparable.

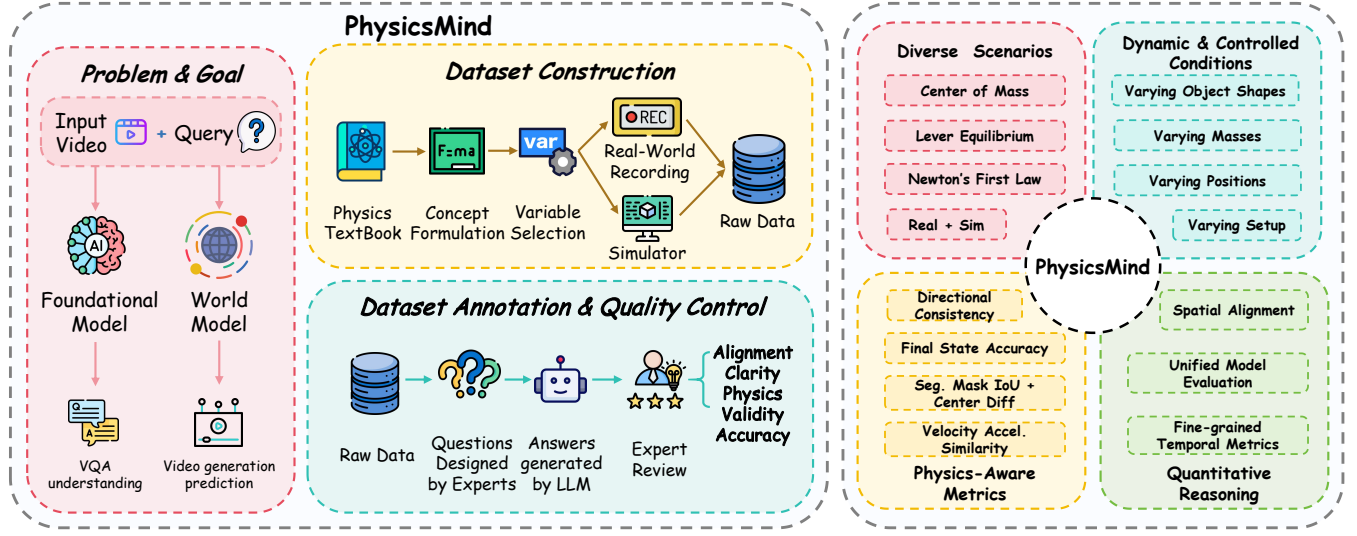


Figure 2: Overview of the PhysicsMind data-construction and evaluation workflow.

3 The PhysicsMind Benchmark

3.1 Problem Formulation

PhysicsMind formulates evaluation in foundation models and world models as a framework spanning VQA and VG. For VQA, the goal is to measure physical reasoning; for VG, the goal is to measure physical consistency under controlled mechanics scenarios, rather than to exhaustively measure general physical intelligence. This shared view enables comparison of error patterns across modalities, including alignment between symbolic answers and trajectory-level judgments and cross-track failure consistency checks.

The evaluation of physical reasoning in foundation and world models can be formalized in a common framework. For foundation models, given an initial frame f and a question q , the model predicts $\hat{a} = f_{\theta}(f, q)$, where $\hat{a} \in \mathcal{A}$; correctness is measured by exact match against the ground truth a^* , i.e., $1[\hat{a} = a^*]$. For world models, the formulation extends to temporal data: given an input video sequence $\mathbf{x} = (x_1, x_2, \dots, x_T)$ and a corresponding physical query q , the model outputs $\hat{y} = g_{\phi}(\mathbf{x}, q)$, where $\hat{y}$ denotes a generated video or derived final-state prediction. Evaluation is scenario-specific via $E_s(\hat{y}, y^*)$, which jointly measures task correctness and physical consistency over transitional dynamics and final outcomes; y^* denotes the ground-truth physically valid target under scenario s . At the dataset level, we report $\mathcal{S} = \mathbb{E}_{(u,s) \sim \mathcal{D}}[E_s(\hat{y}(u), y^*(u))]$, where u is the model input.

3.2 Overview of PhysicsMind

PhysicsMind evaluates physical reasoning in both foundation models and world models through two tracks (left region of Figure 2): Visual Question Answering (VQA) and video generation assessment. The benchmark uses the three mechanics setups in Figure 1, selected to isolate geometric balance, torque-driven causality, and inertia-consistent temporal dynamics, for fine-grained cross-modal

error diagnosis. Each setup is instantiated with real-world recordings and two-dimensional (2D) simulations; where subset pairings are directly comparable, we report targeted sim-to-real transfer.

Controlled data construction is necessary for diagnosing physical reasoning errors because unconstrained scene variation can confound failures of perception, representation, and causal inference. We therefore design each mechanics scenario with systematic, isolated variation in object shapes, masses, initial positions, and setup configurations, so that observed performance changes can be attributed to specific physical factors rather than incidental visual differences. Real-world and simulated results are compared only on directly matched subset pairs, where geometry, dynamics, and labeling protocols are aligned. This protocol is intended to provide robust, scenario-level diagnostics of model behavior under controlled perturbations, not to claim broad, domain-general physical intelligence beyond the benchmark scope.

The dataset comprises short, high-quality video clips with a maximum resolution of 3840×2160 , recorded at 60 FPS and lasting up to 10 seconds. Real-world samples are captured in controlled indoor environments with fixed camera positions to ensure consistent framing and illumination. Simulated samples are generated using a deterministic 2D physics engine, allowing fine-grained control over object attributes and motion reproducibility. Across all scenarios, the dataset incorporates diverse configurations that vary in initial setups, object shapes, masses, and conditions.

Dataset Construction. As shown in the top-middle region of Figure 2, the PhysicsMind dataset is constructed through a structured pipeline that translates visually interpretable and experimentally verifiable laws from high school physics into controlled physical setups. A systematic variable selection strategy is adopted to vary experimental conditions while maintaining control. The dataset includes both simulated and real-world recordings: simulated data is generated using a deterministic 2D physics engine with precise parameter control, while real-world data is captured under

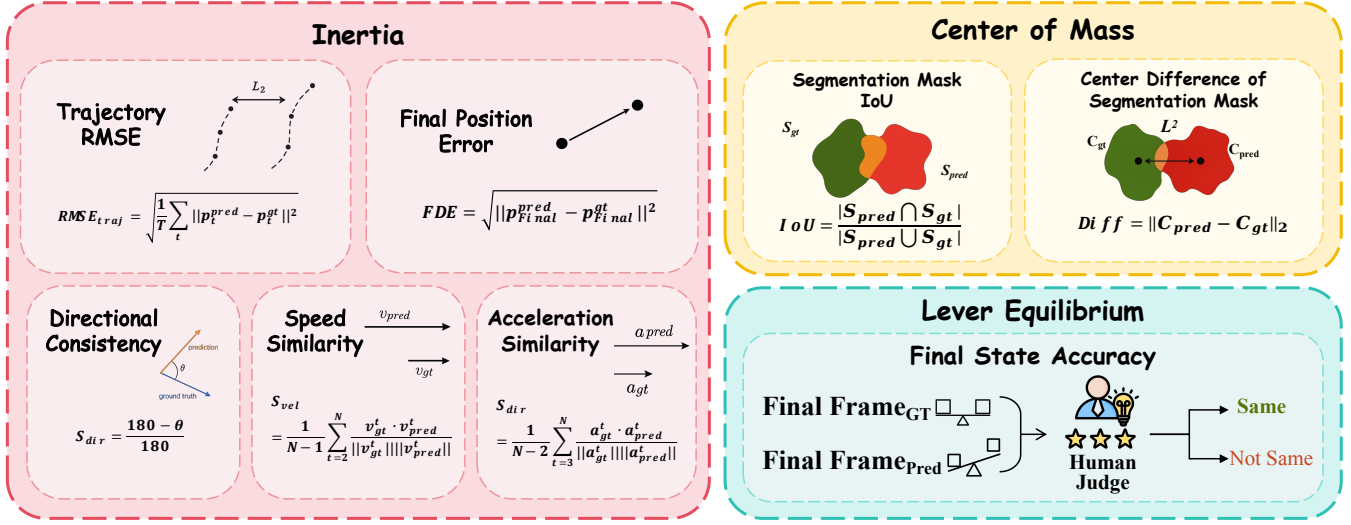


Figure 3: Overview of the video-generation evaluation pipeline in PhysicsMind. The figure maps each mechanics scenario to its corresponding evaluation branch; metric definitions are given in Section 3.3.

fixed camera viewpoints and consistent lighting. Importantly, simulated and real-world data are evaluated in separate splits: 2D simulations provide analytically computed ground-truth quantities evaluated in pixel space, whereas real-world videos adopt a fixed near-orthographic view, enabling image-plane measurements to approximate 3D physical behavior.

Data Annotation and Quality Control. PhysicsMind uses a hybrid annotation workflow that separates human annotation, automatic extraction, and model-based judging (shown in the bottom-middle region of Figure 2). For VQA curation, expert annotators define and manually refine question/answer templates and verify final answer keys for each physics setup, with LLMs used only for assisted template population. For video generation evaluation, human input is targeted rather than exhaustive: annotators mark tracked objects where needed and review a Center-of-Mass sanity-check subset to ensure segmentation-derived centers remain meaningful on generated videos. We then apply the scenario-specific VG evaluation branches consistently across models (Section 3.3) and run manual/automated consistency checks to remove ambiguous, unstable, or physically inconsistent samples.

Benchmark Diversity. PhysicsMind emphasizes controlled variation within each mechanics setup rather than broad domain count. We systematically vary object shapes, masses, and initial positions (Figure 1) and report mechanics-specific metrics across splits. This design supports robust diagnostics of sensitivity to appearance changes and parameter shifts.

3.3 Evaluation Metrics

Because PhysicsMind spans two output modalities and three distinct mechanics scenarios, we adopt scenario-specific metrics rather than collapsing performance into a single score. The three scenarios test different forms of physical consistency: Center of Mass emphasizes geometric consistency in object extent and location, Lever Equilibrium evaluates torque-governed causal balance in

final outcomes, and Newton’s First Law examines temporal inertial consistency during motion. These settings produce different error surfaces across spatial structure, end-state correctness, and trajectory dynamics. A unified aggregate metric would blur these distinctions and can hide failure modes that matter for diagnosis, comparison, and model improvement. The protocol is therefore designed to preserve clear, interpretable error signals.

Video Question Answering. The core metric is binary exact-match accuracy with respect to the ground-truth option. To obtain finer diagnostics, we further report:

- *Category-wise accuracy*, split by mechanics-specific subtypes (e.g., equilibrium vs. adjustment for levers, position vs. rotation for Center of Mass, position vs. stability for inertia), to distinguish different reasoning skills.
- *Experiment-wise accuracy*, aggregated over physical setups and configurations, to reveal which conditions are easier or harder for models.
- *Variant-wise accuracy*, grouped by object type, mass pattern, or motion profile, to test robustness to appearance and parameter changes.

To calibrate model errors against task ambiguity and annotation noise, we also report a human VQA baseline in the main VQA table.

Video Generation. VG evaluation is scenario-specific (Figure 3), and the metric definitions below are the canonical protocol used throughout the paper:

- *Center of Mass.* We compute segmentation-mask IoU and centroid (center-of-mass) deviation between generated and reference outputs using automatically extracted masks (SAM-based pipeline). Because generated-video artifacts can degrade mask quality, we additionally use a human sanity-check subset to verify that segmentation-based evaluation remains a meaningful proxy on generated videos; this subset is used for robustness checking rather than replacing full automatic scoring.

Table 3: VQA Physics Evaluation Results across Models. The Acc columns report accuracy in percentage (%). Values after $\pm$ denote standard deviation across five prompt runs ($n = 5$), reflecting prompt sensitivity rather than example-level uncertainty.

Model	Center of Mass (VQA)			Lever Equilibrium (VQA)			Newton's First Law (VQA)		
	Position Acc (%) $\uparrow$	Rotation Acc (%) $\uparrow$	Overall Acc (%) $\uparrow$	Equilibrium Acc (%) $\uparrow$	Balance Adj. Acc (%) $\uparrow$	Overall Acc (%) $\uparrow$	Obj. Pos. Acc (%) $\uparrow$	Obj. Stability Acc (%) $\uparrow$	Overall Acc (%) $\uparrow$
Human ($n=20$)	95.00 $\pm$ 2.1	92.50 $\pm$ 2.8	93.75 $\pm$ 1.5	93.65 $\pm$ 2.4	90.48 $\pm$ 3.1	92.06 $\pm$ 1.8	96.00 $\pm$ 2.2	95.00 $\pm$ 1.8	95.50 $\pm$ 1.4
Closed-source Reasoning Models									
GPT-5 [30]	60.00 $\pm$ 2.1	80.00 $\pm$ 1.8	70.00 $\pm$ 1.5	66.67 $\pm$ 2.4	85.71 $\pm$ 1.5	76.19 $\pm$ 1.3	60.00 $\pm$ 2.8	95.00 $\pm$ 1.0	77.50 $\pm$ 1.8
o4-mini [31]	50.00 $\pm$ 2.4	55.00 $\pm$ 2.2	52.50 $\pm$ 1.9	61.90 $\pm$ 2.5	52.38 $\pm$ 2.4	57.14 $\pm$ 2.1	75.00 $\pm$ 2.2	95.00 $\pm$ 1.0	85.00 $\pm$ 1.5
GPT-4o [34]	40.00 $\pm$ 2.4	35.00 $\pm$ 2.1	37.50 $\pm$ 1.9	42.86 $\pm$ 2.5	66.67 $\pm$ 2.1	54.76 $\pm$ 1.9	45.00 $\pm$ 2.8	85.00 $\pm$ 1.6	65.00 $\pm$ 2.1
GPT-4-turbo [33]	25.00 $\pm$ 2.2	80.00 $\pm$ 1.8	47.50 $\pm$ 2.0	28.57 $\pm$ 2.3	57.14 $\pm$ 2.2	42.86 $\pm$ 2.0	40.00 $\pm$ 2.5	80.00 $\pm$ 1.8	60.00 $\pm$ 2.0
Claude-4.5-sonnet [4]	45.00 $\pm$ 2.5	20.00 $\pm$ 1.8	32.50 $\pm$ 2.1	61.90 $\pm$ 2.4	61.90 $\pm$ 2.0	61.90 $\pm$ 1.8	10.00 $\pm$ 1.4	90.00 $\pm$ 1.3	50.00 $\pm$ 2.2
Claude-3.7-sonnet [3]	30.00 $\pm$ 2.3	25.00 $\pm$ 1.9	27.50 $\pm$ 1.9	52.38 $\pm$ 2.4	66.67 $\pm$ 2.0	59.52 $\pm$ 1.9	40.00 $\pm$ 2.5	65.00 $\pm$ 2.1	52.50 $\pm$ 2.0
Gemini-2.5-pro [12]	30.00 $\pm$ 2.2	70.00 $\pm$ 1.9	50.00 $\pm$ 1.8	57.14 $\pm$ 2.3	76.19 $\pm$ 1.7	66.67 $\pm$ 1.6	20.00 $\pm$ 1.8	45.00 $\pm$ 2.2	32.50 $\pm$ 1.9
Closed-source Chat Models									
GPT-4o-mini [34]	10.00 $\pm$ 1.5	20.00 $\pm$ 1.8	15.00 $\pm$ 1.6	23.81 $\pm$ 2.1	38.10 $\pm$ 2.2	30.95 $\pm$ 1.8	10.00 $\pm$ 1.4	80.00 $\pm$ 1.8	45.00 $\pm$ 2.1
GPT-4.1 [33]	21.30 $\pm$ 2.2	46.75 $\pm$ 2.3	43.00 $\pm$ 2.0	50.50 $\pm$ 2.5	55.25 $\pm$ 2.2	37.75 $\pm$ 2.1	47.25 $\pm$ 2.6	41.03 $\pm$ 2.3	47.60 $\pm$ 2.0
Claude-3.5-sonnet [2]	25.00 $\pm$ 2.2	20.00 $\pm$ 1.8	22.50 $\pm$ 1.9	19.05 $\pm$ 2.0	52.38 $\pm$ 2.2	35.71 $\pm$ 1.9	40.00 $\pm$ 2.5	90.00 $\pm$ 1.3	65.00 $\pm$ 2.0
Gemini-2.5-flash [12]	15.00 $\pm$ 1.8	55.00 $\pm$ 2.2	35.00 $\pm$ 1.9	52.38 $\pm$ 2.4	61.90 $\pm$ 2.0	57.14 $\pm$ 1.8	40.00 $\pm$ 2.5	85.00 $\pm$ 1.6	62.50 $\pm$ 1.9
Gemini-2.5-flash-image [12]	25.00 $\pm$ 2.0	40.00 $\pm$ 2.1	32.50 $\pm$ 1.8	42.86 $\pm$ 2.4	61.90 $\pm$ 2.0	52.38 $\pm$ 1.9	45.00 $\pm$ 2.6	90.00 $\pm$ 1.3	67.50 $\pm$ 1.8
Open-source Reasoning Models									
Qwen3-vl-8b-instruct [39]	25.00 $\pm$ 2.0	40.00 $\pm$ 2.1	32.50 $\pm$ 1.9	38.10 $\pm$ 2.3	28.57 $\pm$ 2.0	33.33 $\pm$ 1.8	40.00 $\pm$ 2.5	95.00 $\pm$ 1.0	67.50 $\pm$ 1.9
Qwen-vl-max [38]	25.00 $\pm$ 2.0	30.00 $\pm$ 1.9	27.50 $\pm$ 1.8	52.38 $\pm$ 2.4	57.14 $\pm$ 2.1	54.76 $\pm$ 1.9	40.00 $\pm$ 2.5	85.00 $\pm$ 1.6	62.50 $\pm$ 1.9
Qwen2.5-vl-72b-instruct [35]	30.00 $\pm$ 2.1	30.00 $\pm$ 1.9	30.00 $\pm$ 1.7	47.62 $\pm$ 2.4	57.14 $\pm$ 2.0	52.38 $\pm$ 1.8	40.00 $\pm$ 2.5	85.00 $\pm$ 1.6	62.50 $\pm$ 1.9
Deepseek-vl2 [43]	40.00 $\pm$ 2.4	70.00 $\pm$ 2.0	55.00 $\pm$ 1.8	42.86 $\pm$ 2.3	23.81 $\pm$ 1.9	33.33 $\pm$ 1.7	10.00 $\pm$ 1.3	95.00 $\pm$ 1.0	52.50 $\pm$ 2.0
Open-source Chat Models									
Qwen2.5-vl-32b-instruct [39]	40.00 $\pm$ 2.4	60.00 $\pm$ 2.1	50.00 $\pm$ 1.9	23.81 $\pm$ 2.0	47.62 $\pm$ 2.1	35.71 $\pm$ 1.8	40.00 $\pm$ 2.5	75.00 $\pm$ 1.9	57.50 $\pm$ 2.0
Llama-3.2-90b-vision-instruct [16]	55.00 $\pm$ 2.5	55.00 $\pm$ 2.2	55.00 $\pm$ 1.9	42.86 $\pm$ 2.3	61.90 $\pm$ 2.0	52.38 $\pm$ 1.8	35.00 $\pm$ 2.4	95.00 $\pm$ 1.0	65.00 $\pm$ 1.9
Llama-3.2-11b-vision-instruct [16]	20.00 $\pm$ 1.8	50.00 $\pm$ 2.2	35.00 $\pm$ 1.9	33.33 $\pm$ 2.2	28.57 $\pm$ 2.0	30.95 $\pm$ 1.7	15.00 $\pm$ 1.6	80.00 $\pm$ 1.8	47.50 $\pm$ 2.0
Grok-4 [44]	25.00 $\pm$ 2.0	60.00 $\pm$ 2.2	42.50 $\pm$ 1.9	47.62 $\pm$ 2.3	28.57 $\pm$ 2.0	38.10 $\pm$ 1.8	75.00 $\pm$ 2.2	80.00 $\pm$ 1.6	77.50 $\pm$ 1.7
Glm-4.5V [37]	35.00 $\pm$ 2.2	35.00 $\pm$ 2.0	35.00 $\pm$ 1.8	61.90 $\pm$ 2.3	28.57 $\pm$ 2.0	45.24 $\pm$ 2.0	60.00 $\pm$ 2.5	75.00 $\pm$ 1.9	67.50 $\pm$ 1.8

- *Lever Equilibrium.* We evaluate final-state correctness using a model judge (GPT-5.1) on the generated final frame, checking agreement with the expected equilibrium outcome. This is a final-state agreement protocol and is intentionally separated from full temporal trajectory analysis (see dedicated subsection below).
- *Newton's first law.* We adopt a semi-automatic trajectory protocol: humans annotate the object of interest, CoTracker extracts trajectories over frames, and we compute trajectory-RMSE, final-position error, speed similarity, acceleration similarity, and directional consistency to test inertia-consistent motion.

Detailed formulations are provided in the Appendix.

4 Experiments

Our study is guided by the following research questions:

- **RQ1. Physical Reasoning in Perception:** Can large multimodal foundation models reason about physical laws from static visual inputs?
- **RQ2. Physical Consistency in Generation:** Do world or video generation models produce motion that remains consistent with core physical principles?
- **RQ3. Sim-to-Real Transfer:** To what extent do model behaviors transfer between simulated and real-world settings

under our controlled mechanics scenarios, and how does this evidence vary by scenario?

Across these three questions, we focus on controlled mechanics diagnostics rather than broad claims about general physical intelligence. The results are used to highlight recurring failure modes (e.g., appearance shortcuts, uneven subtype reasoning, and mixed sim-to-real transfer) under this benchmark setup.

4.1 Experimental Setup and Evaluation Protocol

All models are evaluated with the fixed splits and protocols defined in Section 3.3. In VQA, each model receives an image and a multiple-choice question; we report exact-match accuracy with category/variant breakdowns and include a human baseline for calibration. For VG, we run the designated evaluation branch (SAM-based mask pipeline, GPT-5.1 final-frame judge, or human-seeded CoTracker tracking), apply the same scripts across models, and report mean $\pm$ std over $n = 5$ prompt runs.

4.2 Evaluation of Physical Reasoning in VQA (RQ1)

Table 3 summarizes VQA results for multimodal foundation models, addressing RQ1: how reliably contemporary vision-language systems apply mechanics-consistent reasoning under controlled visual tests. Because this benchmark is intentionally compact and

Table 4: Video generation physics evaluation under controlled PhysicsMind scenarios. Columns report physical-consistency metrics (mean $\pm$ std over $n = 5$ prompt runs); these deviations reflect prompt-run variation rather than example-level confidence. The best value in each column is bold.

Model	Center of Mass (Video)		Lever Equilibrium (Video)	Newton’s First Law (Video)				
	Seg. Mask IoU $\uparrow$	Seg. Mask Center $\downarrow$	Final State Acc. (%) $\uparrow$	Trajectory RMSE $\downarrow$	Final. Position Error $\downarrow$	Speed Similarity $\uparrow$	Acceleration Similarity $\uparrow$	Directional Consistency $\uparrow$
Veo3.1 [15]	0.019 $\pm$ 0.003	108.39 $\pm$ 12.5	35 $\pm$ 6.2	0.384 $\pm$ 0.012	0.198 $\pm$ 0.009	-0.011 $\pm$ 0.008	-0.021 $\pm$ 0.007	0.5419 $\pm$ 0.014
Sora-2 [32]	0.167 $\pm$ 0.008	121.42 $\pm$ 11.8	40 $\pm$ 6.5	0.380 $\pm$ 0.011	0.199 $\pm$ 0.008	-0.042 $\pm$ 0.009	0.017 $\pm$ 0.006	0.5494 $\pm$ 0.013
LTX-Video [18]	0.005 $\pm$ 0.001	76.37 $\pm$ 9.2	4.76 $\pm$ 2.1	0.406 $\pm$ 0.013	0.213 $\pm$ 0.010	-0.011 $\pm$ 0.008	-0.056 $\pm$ 0.009	0.5594 $\pm$ 0.015
CogVideoX1.5-5B-I2V [45]	0.014 $\pm$ 0.002	223.70 $\pm$ 18.5	38.10 $\pm$ 6.8	0.414 $\pm$ 0.014	0.323 $\pm$ 0.012	0.090 $\pm$ 0.010	-0.043 $\pm$ 0.008	0.4884 $\pm$ 0.016
Pyramid Flow [23]	0.012 $\pm$ 0.002	322.97 $\pm$ 22.3	47.62 $\pm$ 7.2	0.381 $\pm$ 0.011	0.276 $\pm$ 0.011	-0.047 $\pm$ 0.009	-0.019 $\pm$ 0.007	0.6437 $\pm$ 0.012
Wan2.2 14B [40]	0.136 $\pm$ 0.007	181.39 $\pm$ 15.2	33.3 $\pm$ 6.0	0.395 $\pm$ 0.012	0.134 $\pm$ 0.007	0.014 $\pm$ 0.006	-0.098 $\pm$ 0.010	0.4775 $\pm$ 0.015
Cosmos-predict2 2B [1]	0.009 $\pm$ 0.001	217.33 $\pm$ 17.8	42.85 $\pm$ 6.7	0.350 $\pm$ 0.010	0.243 $\pm$ 0.010	-0.013 $\pm$ 0.008	-0.019 $\pm$ 0.007	0.4884 $\pm$ 0.016

controlled, we emphasize recurring patterns and error types over fine-grained ranking from small score differences. A human VQA baseline is included as a calibration reference for task difficulty, while the discussion remains benchmark-focused on cross-model patterns and failure modes.

Current models can often answer some items correctly, but they do not sustain consistent performance across controlled mechanics settings. The error profile is also uneven across subtypes within each scenario, with gains on easier or more visually salient questions not transferring to harder variants that require coordinated geometric and causal judgments. This pattern points to fragmented competence rather than a unified mechanics understanding. In observed failures, predictions are frequently driven by superficial appearance cues while missing key relational constraints needed to infer stable outcomes. We therefore interpret the current VQA results as evidence of partial, brittle physical reasoning under controlled evaluation.

Overall Performance Trends. Across all models, overall accuracy ranges from roughly random guessing (25%) up to 85%, revealing large disparities in physical reasoning competence. No single model dominates across all three domains. GPT-5 reaches the highest overall scores on Center of Mass and Lever Equilibrium (70% overall in both), whereas o4-min attains the highest Newton’s First Law accuracy (85%). Given the benchmark scale and prompt-run variability, we avoid over-interpreting close model-to-model gaps and instead focus on these repeated cross-domain inconsistencies.

Domain-Specific Insights. Performance differs markedly by physical domain. Newton’s First Law questions obtain the highest mean accuracy (68.5% on average), which may partly reflect visually salient motion cues and relatively simple binary decisions (move vs. remain still). Center of Mass questions average 47.3%; these items require implicit localization of equilibrium axes and reasoning about hidden mass distributions, which remain challenging for most models. Lever Equilibrium questions lie in between (mean 52.9%), indicating partial but uneven torque reasoning when textual mass annotations are available.

Error Characterization. Qualitative inspection (Figure 4) reveals two dominant error modes that appear across models and domains. First, models frequently misread fine-grained visual details such as small mass labels or lever markers, leading to incorrect torque comparisons (*visual parsing errors*). Second, they often fail to complete

the full reasoning chain from perception to physics computation and textual inference, producing answers that are locally plausible but globally inconsistent with mechanics (*incomplete reasoning*). For example, some models compare left and right object masses but ignore whether the objects are actually placed on the lever, yielding logically invalid torque calculations. These patterns expose a gap between visual perception and multi-step, physically grounded reasoning.

Interpretation. Overall, the results indicate that large multimodal models can capture coarse spatial regularities (e.g., heavier objects tend to lower the lever) but still struggle with precise quantitative relationships and counterfactual reasoning. Top-performing systems exceed random-choice baselines in several settings, yet persistent domain gaps and systematic error modes indicate limited physical reasoning under this evaluation setup. PhysicsMind therefore serves as a diagnostic probe of current limitations in static and short-horizon visual physics reasoning.

4.3 Evaluation of Physical Consistency in Video Generation (RQ2)

Table 4 reports quantitative results for seven recent video generation models under the predefined mechanics subsets. This section addresses **RQ2**, asking whether current world models produce motion that remains consistent with core mechanical principles. As in RQ1, we emphasize stable qualitative patterns and failure modes over fine-grained ranking from small metric differences.

Overall Performance Trends. Across all experiments, physical consistency remains uneven. No model achieves uniformly strong performance across spatial (Center of Mass), mechanical (torque balance), and dynamic (motion prediction) metrics. For Center of Mass, models such as Sora-2 and Wan2.2 14B obtain the highest segmentation IoU, whereas Pyramid Flow yields the best lever final-state accuracy (47.62%). Under Newton’s First Law, Cosmos-predict2 2B achieves the lowest trajectory error (normalized RMSE = 0.350), yet its directional consistency is still moderate rather than near-perfect. These mixed patterns suggest that existing generators can produce visually plausible videos while still struggling to maintain physically consistent dynamics.

Domain-Specific Insights. *Center of Mass.* Most models still miss precise geometry and centroids. IoU remains low (10^{-2} – 10^{-1}) and

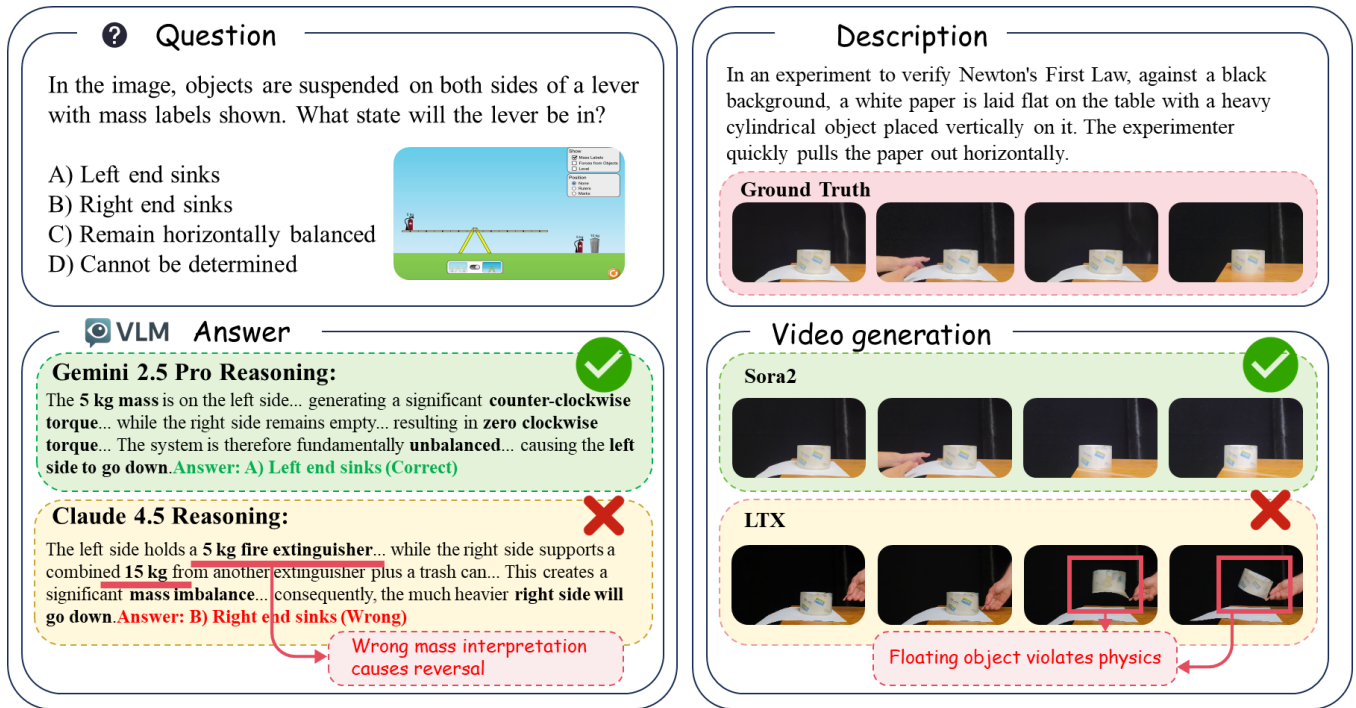


Figure 4: Qualitative error examples for visual reasoning and video generation. Left: one lever-equilibrium case where Gemini 2.5 Pro predicts the correct outcome while Claude 4.5 does not. Right: one Newton’s First Law case where Sora-2 better preserves inertia-consistent motion, whereas LTX-Video shows implausible lift or acceleration.

centroid errors stay high (about 75–320+ pixels), indicating coarse placement rather than reliable equilibrium structure.

Lever Equilibrium. Mechanical reasoning around torque balance remains limited. Most models cluster near the 50% random baseline (e.g., Sora-2: 40%, Pyramid Flow: 47.62%), and some fall well below chance (LTX-Video: 4.76%). Under this evaluation setup, this pattern is more consistent with appearance-driven heuristics than with stable torque reasoning.

Newton’s First Law. Kinematic stability is also limited. While Cosmos-predict2 2B attains the smallest trajectory error, directional consistency values cluster around 0.5–0.65 (where 0.5 corresponds to random direction alignment), indicating only partial preservation of velocity direction. Together with weak speed and acceleration similarity for several models, these results suggest frequent deviations from inertia-consistent trajectories.

Error Characterization. Qualitative inspection (Figure 4, right) highlights systematic temporal reasoning failures. In some cases (e.g., Sora-2), objects remain approximately at rest when the supporting paper is pulled quickly, aligning with Newton’s First Law; in others (e.g., LTX-Video), objects lift or accelerate in implausible ways. Similar to the VQA setting, these errors arise from inadequate object-state tracking and missing causal links between external forces and motion response, with models often treating frames as nearly independent images instead of propagating coherent velocity over time.

Interpretation. Current video generation systems can exhibit perceptual realism without reliably preserving physical consistency

under our mechanics probes. They appear to rely heavily on visual priors, yielding sequences that look convincing but still often violate balance, torque, and inertia metrics. Bridging this gap likely requires world models that better encode conservation-aware dynamics across both space and time, rather than relying only on large-scale text–image–video pretraining.

4.4 Sim-to-Real Transfer Under Controlled Mechanics (RQ3)

RQ3 measures sim-to-real transfer on matched simulated/real subsets using signed gaps in Table 5 (**Gap = Simulated – Real**). Because subset matching and metric definitions are scenario- and modality-specific, we treat these results as controlled diagnostics within this benchmark rather than evidence of domain-invariant transfer.

VQA. VQA gaps are scenario-dependent rather than uniform. In Center of Mass, overall gaps are negative for five models (–35.00 to –15.00) and mildly positive for three (2.50 to 7.50). Lever Equilibrium remains mixed around zero (–9.52 to 5.96). In contrast, Newton’s First Law overall gaps are positive for all models (0.13 to 12.20), although subtype gaps can differ in sign within the same model. This cross-scenario sign shift indicates uneven transfer behavior in the matched VQA subsets.

Video generation. Video-generation transfer follows a different pattern. For Lever Equilibrium, six of seven models are simulation-favoring (Final-State Gap 2.31 to 13.81), with Sora-2 as the exception

Table 5: Sim2Real transfer summary under controlled mechanics. Top: VQA sim-to-real gaps across three mechanics scenarios. Bottom: -generation sim-to-real gaps across the same scenarios. In all cases, Gap = Simulated – Real. Values closer to 0 indicate smaller sim-to-real discrepancy, while the sign indicates the direction of the domain difference.

(a) VQA sim-to-real transfer summary

Model	Center of Mass			Lever Equilibrium			Newton's First Law		
	Position Gap	Rotation Gap	Overall	Equilibrium Gap	Balance Adj. Gap	Overall	Obj. Pos. Gap	Obj. Stability Gap	Overall
GPT-5	-55.00±4.5	-15.00±2.6	-35.00±3.8	-9.53±2.5	-7.14±3.4	-8.34±3.3	+4.62±3.6	-1.83±2.0	+1.40±2.5
o4-min	-50.00±3.3	0.00±1.8	-25.00±4.2	-9.52±2.2	-9.52±2.5	-9.52±3.2	+5.17±3.0	+2.94±2.2	+4.06±3.1
Claude-sonnet-4-5	-15.00±4.1	-15.00±2.5	-15.00±4.1	-11.90±3.9	+2.39±2.4	-4.76±2.4	-4.31±3.7	+4.56±2.7	+0.13±2.0
Gemini-2.5-pro	-45.00±3.4	+10.00±4.2	-17.50±3.7	-7.14±3.5	-4.76±3.4	-5.95±3.0	+2.83±3.4	+4.17±2.8	+3.50±3.1
Grok-4	-35.00±4.9	+5.00±3.2	-15.00±4.2	+2.38±2.7	+9.53±3.3	+5.96±2.2	+5.48±2.5	+3.72±2.6	+4.60±2.3
Qwen3-vl-8b-instruct	0.00±1.8	+15.00±2.7	+7.50±2.6	-9.53±3.2	+4.76±2.8	-2.39±2.4	+14.83±2.9	+9.57±2.6	+12.20±4.4
Deepseek-vl2	-5.00±3.2	+10.00±3.7	+2.50±2.1	-4.76±3.4	+9.52±2.4	+2.38±2.4	+9.72±3.8	+14.31±3.8	+12.02±3.6
Glm-4.5v	-5.00±3.0	+10.00±3.3	+2.50±2.4	-14.28±3.0	-4.76±3.0	-9.52±2.6	+13.45±3.2	+10.28±3.6	+11.87±3.8

(b) Video-generation sim-to-real transfer summary

Model	Center of Mass		Lever Equilibrium	Newton's First Law				
	IoU Gap	Center Gap	Final-State Gap	Traj. RMSE Gap	Final Pos. Gap	Speed Sim. Gap	Accel. Sim. Gap	Dir. Cons. Gap
Veo3.1	+0.1332±0.0056	-4.11±1.7	+3.27±2.3	-0.032±0.0055	-0.054±0.0055	+0.038±0.0065	+0.027±0.0054	+0.0429±0.0046
Sora-2	-0.0866±0.0053	-22.07±1.8	-4.17±1.7	+0.018±0.0121	+0.031±0.0073	-0.025±0.0118	-0.012±0.0104	-0.0193±0.0063
LTX-Video	+0.0226±0.003	+25.61±1.3	-3.81±1.4	-0.041±0.0103	-0.072±0.0141	+0.052±0.0074	+0.039±0.0042	+0.0614±0.0135
CogVideoX1.5-5B-I2V	+0.0574±0.0055	-16.53±1.0	+2.31±1.5	-0.027±0.006	-0.098±0.0082	+0.061±0.0128	+0.034±0.0088	+0.0347±0.0043
Pyramid Flow	+0.0489±0.003	-191.52±2.1	+11.43±1.7	-0.035±0.0134	-0.064±0.0124	+0.044±0.0053	+0.022±0.0127	+0.0782±0.0096
Wan2.2 14B	+0.0479±0.0036	-90.91±1.0	+8.73±2.2	-0.022±0.006	-0.042±0.0072	+0.033±0.009	+0.048±0.0141	+0.0265±0.0112
Cosmos-predict2 2B	+0.1588±0.0061	-11.16±1.4	+11.90±1.9	-0.029±0.0099	-0.057±0.004	+0.047±0.0122	+0.031±0.0054	+0.0456±0.0094

(-4.17). For Newton's First Law, the same six models are simulation-favoring across all five metrics (negative Trajectory/Final-Position error gaps and positive Speed/Acceleration/Direction similarity gaps), while Sora-2 reverses this direction on all five metrics. Center-of-Mass results are less coherent: IoU favors simulation for six models, but centroid gaps are mixed, including a real-favoring reversal for LTX-Video (25.61).

Synthesis. Table 5 supports a conservative conclusion: sim-to-real transfer is modality- and scenario-dependent in these matched subsets. VQA does not show a single global gap direction, while video generation is mostly simulation-favoring with a clear model-level exception. We therefore interpret RQ3 as controlled evidence of domain sensitivity under this benchmark, not as broad sim-to-real generalization.

4.5 Discussion and Implications

Across RQ1–RQ3, models exploit regularities more than they preserve mechanics-consistent structure. In VQA, this yields uneven subtype performance and inconsistent answers. In video generation, sequences violate mechanics-consistent dynamics through center-of-mass, inertia, and temporal-continuity failures. In sim-to-real analysis, brittleness appears as scenario- and modality-dependent gaps. PhysicsMind indicates competence with fragile causal integration, not law-consistent reasoning.

The distinction between discriminative and generative evaluation is central. Discriminative physical reasoning (VQA) tests whether a model maps a configuration to a correct choice and can benefit from coarse visual cues even when the reasoning chain is incomplete. Generative physical consistency (VG) is stricter: the model must maintain coherent state evolution over time, so weak temporal propagation and missing causal constraints are harder to

hide. Thus, strong option selection does not guarantee physically consistent dynamics.

Although PhysicsMind does not span all dimensions of physical intelligence, its controlled mechanics design offers diagnostic value that broader multimedia benchmarks can dilute. By reducing incidental variation and using matched simulated/real subsets, it isolates weaknesses: appearance-driven shortcuts, incomplete causal reasoning, weak temporal consistency, and non-uniform sim-to-real transfer. The benchmark complements multimedia benchmarks by indicating whether gains reflect deeper mechanics understanding or improved surface-level matching under controlled cross-modality diagnostic settings.

5 Conclusion

PhysicsMind provides a controlled testbed for examining how modern foundation and world models connect perception, reasoning, and prediction to concrete mechanics laws rather than visual plausibility alone. Across three textbook scenarios with paired simulated and real videos, we find that both vision-language and video generation models still rely heavily on superficial cues and frequently violate basic constraints on balance, torque, and inertia. These consistent failures across modalities indicate that robust, law-consistent physical understanding has not yet emerged under controlled evaluation. The benchmark is intentionally bounded: it currently focuses on rigid-body setups, short time horizons, and off-the-shelf models. Extending the benchmark to richer physical phenomena (e.g., friction, collisions, fluids), longer and interactive scenarios, and training or architecture design for physics-aware models remains important future work. The modular pipeline is designed to support these extensions while preserving controlled, interpretable diagnostics.

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